

What Determines a Simulation Benchmark Ranking? Structural Blindness, Finite Configuration Grids, and Evaluation Budget in Simulation-Based Policy Comparison

Jun Ji¹, Bowen Tan², Yi Li¹, Xiaolei Zhang¹, Shengjie Guo³, Yi Sui^{1*}

^{1*}College of Computer Science and Technology, Qingdao University, Qingdao, 266071, China.

²The Hong Kong University of Science and Technology, Hong Kong SAR, China.

³Inner Mongolia Agricultural University, Hohhot, China.

*Corresponding author(s). E-mail(s): suiyi@qdu.edu.cn;

Contributing authors: junji@qdu.edu.cn; btanab@connect.ust.hk; ly2005@qdu.edu.cn;
zhangxiaolei@qdu.edu.cn; guosj@emails.imau.edu.cn;

Abstract

Simulation benchmarks are increasingly used to rank robot foundation policies, treating a calibrated simulator as evidence about which is better. We ask a question prior to how well simulation correlates with reality: given the evidence used to build and run such a benchmark, which ranking claims does that evidence determine? We answer on a widely used sim-to-real suite, across two embodiments, enumerating every task’s configuration grid. First, the calibration protocol is structurally blind: replaying 98 real demonstrations identifies the joint-controller ratio and the execution delay, but a common gain scaling and the torque limit leave the replay trajectory unchanged, by 0.14–0.21 mm and bitwise exactly, on two independent stacks. Second, the benchmark is a finite configuration population rather than a sample: the episode index fixes the initial state modulo 24 to 300 configurations, two widely used ports enumerate different grids, and enumerating the grid, which zeroes configuration sampling error, leaves the suite’s unreported averaging over four robot texture variants, spanning 0.400 in success rate, as the largest term in the uncertainty budget. Third, following the published protocol reproduces its rates to 0.026 mean absolute difference, yet that same 72-episode budget resolves only one of the four orderings it reports, and a second reverses sign under faithful re-run. We then measure what a set-valued verdict over the compatible parameters buys, and where it stops: on the one pair whose published real-robot results contradict the simulator, it declares the ordering the real robot reverses. Three of our own claims are retracted.

Keywords: Policy evaluation, Sim-to-real transfer, Identifiability, Partial identification, Benchmark reproducibility, Robot manipulation

1 Introduction

A simulator calibrated against real demonstrations is increasingly treated as evidence about

which robot policy is better. Simulation suites that mirror real evaluation scenes (Li et al., 2025) are used to compare foundation policies at a scale

no physical laboratory can match, and their published tables are read as rankings: this policy beats that one on this task. The appeal is obvious. A physical evaluation of a single policy on a single task costs hours of robot time and is difficult to reproduce; a simulated one costs minutes and is, in principle, exactly repeatable.

The standard question asked of such a suite is how well it correlates with reality, and it is usually answered with a correlation coefficient or a rank agreement over a handful of policies. This paper asks a question that comes before that one, and that can be answered without a robot:

Given the evidence actually used to build and run a simulation benchmark — the demonstrations it was calibrated on, the configurations it enumerates, and the episodes it runs — which ranking claims does that evidence determine?

The distinction matters because a correlation study treats the simulator’s output as a measurement with some error attached, whereas our question asks whether the output is a well-defined function of the inputs at all. If two simulator configurations are equally consistent with every piece of calibration data and they disagree about which policy wins, then no amount of correlation analysis repairs the ranking: the evidence does not contain the answer. This is the distinction between *identifiability* and *statistical power*, and one of our findings is that practitioners systematically conflate them — including, at two points in this project’s history, us.

We answer the question on a widely used sim-to-real evaluation suite for manipulation, across two embodiments (a WidowX bridge setup and a Google-robot fractal setup), eight policy configurations across three model families — Octo (Ghosh et al., 2024), OpenVLA (Kim et al., 2025) and RT-1 (Brohan et al., 2023) — and a complete enumeration of every task’s configuration grid. Three findings, each independent of the others.

The calibration protocol is structurally blind (Sect. 4).

The protocol identifies some dynamics parameters and is *exactly* uninformative about others. Replaying 98 real demonstrations, we show analytically and then empirically that a common scaling of the joint PD gains and the value of the torque limit leave the replay trajectory unchanged — the

first because the free-space response depends only on the ratio d/k , the second because commanded torque never saturates along demonstrated velocities. The measured effect is 0.141 mm on one simulator stack and 0.207 mm on a second, independent one for the gain scaling, and bitwise zero for the torque limit. These are structural invariances of the protocol, not flat regions of a noisy objective, so no improvement in fitting technique removes them.

The benchmark is a finite configuration population, not a sample (Sect. 5).

The episode index determines the initial state modulo the size of the task’s configuration grid — 24 configurations for three bridge tasks, 64 for one port’s eggplant task, 300 for the fractal coke-can task. Episodes beyond that count are exact repeats, so the common practice of running more episodes stops buying information about configurations at a threshold most users never learn. Two widely used ports of the same suite enumerate *different* grids, so a number computed on one is not comparable to a number computed on the other. Enumerating the grid drives configuration sampling error to exactly zero, and what remains is instructive: policy-seed noise, implementation-build drift, and — larger than either — the suite’s own unreported averaging over four robot *texture* variants, across which one policy’s success rate ranges over 0.400. That last parameter changes the colour of the robot model, carries no physics whatsoever, and is invisible to replay calibration for exactly that reason.

The published protocol does not resolve the rankings it reports (Sect. 6).

We follow the reference protocol exactly and reproduce the published success rates to a mean absolute difference of 0.026 across eight cells, and on the second embodiment to +0.003 and −0.040 — so the pipeline is faithful, which is what licenses the rest. At that same 72-episode budget, only one of the four policy orderings the protocol reports is statistically resolvable, and on one task a faithful re-run of the identical protocol *reverses the published sign*. Both values sit far inside the seed noise, so this is not a contradiction; it is the same non-resolvable comparison resolved differently by

chance twice, which is precisely the failure mode this paper is about.

Having established that ranking claims are underdetermined from three independent directions, we ask what can be done about it (Sects. 7 and 8). The natural remedy is a set-valued verdict: declare an ordering only when it holds for every simulator parameter the calibration data cannot exclude. We measure that criterion’s reach, and report both halves of the answer. It is conservative as designed on synthetic ground truth, where the point criterion’s error rate *rises* with episode count to 0.88, and it stabilises verdicts on real replicate data. At the one- to three-seed budgets the literature actually uses it disagrees with point calibration on 3 of 17 policy pairs — but when we spend more seeds, two of those three disagreements turn out to have been our own insufficient power, leaving one. And on the single policy pair for which published real-robot results contradict the simulator, the criterion over the dynamics compatible set *declares the ordering the real robot reverses*. Its reach stops at the parameter family the calibration data can constrain, and the sim-to-real gap on that pair lies outside it.

1.1 Contributions

1. (Sect. 4) An exact structural non-identifiability in demonstration-replay calibration, established analytically and confirmed on two independent simulator stacks to sub-millimetre agreement.
2. (Sect. 5) The identification of the benchmark as a finite configuration population, with the configuration count per task, the demonstration that two widely used ports evaluate different benchmarks, and the consequences for what more episodes can buy.
3. (Sect. 5) An estimand taxonomy and an uncertainty budget in which the largest measured term is a visual variant the benchmark averages over without reporting it.
4. (Sect. 6) A faithful reproduction of the published protocol showing that its own budget resolves one of the four orderings it reports, and that one ordering reverses under re-run.
5. (Sect. 7) A decision-relevant instance of non-identifiability — an exactly invisible torque limit that displaces a policy difference by 0.121

and moves verdicts from decidable to abstaining — stated, after our own larger-budget replication, as a property of the evaluation budget rather than of parameter sign.

6. (Sect. 8) A verdict ladder with its coverage and power measured on synthetic ground truth and its stability measured on real replicates, together with the negative result that delimits it: on the one pair with contradicting real-robot ground truth, the criterion declares the wrong ordering.
7. (Appendices) Reproducibility assets, including the original reference stack running headless on a GPU-less host through a purpose-built Vulkan compatibility layer, the configuration-census protocol, and append-only provenance logs for every episode reported here.

1.2 What we withdraw

Three conclusions of our own are retracted in this paper, each after we ran the experiment designed to test it.

1. “*Eggplant nominal: octo-small beats octo-base.*” Declared from one seed set with a per-episode bootstrap, interval $[+0.08, +0.32]$. Three same-generation seed sets give $+0.141$, -0.031 and -0.047 . Withdrawn (Sect. 5.5).
2. “*The torque effect replicates across two seed sets.*” The episode index fully determines the configuration, and the policy in question decodes deterministically, so the two “seed sets” were repeat runs of identical rollouts. Withdrawn and replaced by a configuration-census protocol (Sect. 5.3).
3. “*Budget cannot explain the point-versus-set disagreements.*” Taking the census from three to five seed sets resolved two of the three disagreements, and dissolved the one between-condition sign reversal we had claimed. Withdrawn; Sect. 7 states the weaker conclusion the data supports.

Four further ranking-flip candidates seen at 24 and 48 episodes were withdrawn before being claimed, when adding episodes removed them. We list this in the body rather than in a limitations section because a reader’s estimate of how much to trust our surviving claims should be calibrated by how the non-surviving ones died.

1.3 Scope

We make no claim about which policy is actually better on any real robot, and we ran no hardware ourselves. Where real-robot ground truth is needed — in Sect. 8.4’s delimiting experiment — we use the success rates published alongside the benchmark for the same policies and task, which are the reference values the benchmark itself is validated against. Two public evaluation services that would have provided independent hardware results were unavailable during this work: one endpoint was offline throughout and the other’s queue allocation could not be confirmed. Our claims are about what the simulated evidence determines, which is a question about the benchmark rather than about the robot.

2 Related work

Simulation-based evaluation of manipulation policies.

Recent evaluation suites construct simulated scenes that mirror real evaluation setups, calibrate the simulator against real demonstrations, and report success rates intended to be comparable with real-robot numbers (Li et al., 2025). Parallel efforts pursue automated real-robot evaluation (Zhou, Pertsch, Levine, et al., 2025) and distributed evaluation across laboratories (Atreya et al., 2025). All are motivated by the cost and irreproducibility of physical evaluation, and all are validated by some form of sim-to-real agreement: rank correlation, mean absolute difference, or per-task scatter against real rates. Our work takes such a suite as its object of study rather than as a tool, and asks not how closely it agrees with reality but which of its own outputs are determined by its own inputs.

Identifiability and system identification.

That not every parameter of a dynamical model can be recovered from a given excitation is classical, and structural identifiability analysis is standard in system identification (Bellman & Åström, 1970; Ljung, 1999). Robot calibration practice generally acknowledges it in the form of excitation design: trajectories are chosen to make the parameters of interest observable (Swevers, Ganseman, Tükel, De Schutter, & Van Brussel, 1997). The

gap we address is that demonstration-replay calibration does not get to choose its excitation — the demonstrations are whatever humans happened to teleoperate, they are free-space and slow, and the resulting blindness is inherited rather than designed. We treat the blind directions not as a nuisance to be reduced but as a set to be propagated, which connects to the next literature.

Partial identification and set-valued inference.

When data do not pin down a parameter, one can report the set of values consistent with the data and the corresponding set of implied conclusions, rather than a point estimate and a confidence interval around it (Manski, 2003; Tamer, 2010). Model-confidence-set procedures provide the specific device we use: invert a test to obtain the collection of models the data cannot distinguish, then require a conclusion to hold across that collection (Hansen, Lunde, & Nason, 2011). Our contribution is not methodological novelty here but instantiation and measurement — we construct the compatible set for a real calibration protocol, propagate it to policy rankings, and quantify what the resulting criterion does and does not buy, including a case where it fails.

Reproducibility and statistical practice in learned-policy evaluation.

Sensitivity of reinforcement learning results to random seeds, and the resulting unreliability of small-sample comparisons, is documented and has produced concrete recommendations on reporting (Agarwal, Schwarzer, Castro, Courville, & Bellemare, 2021; Colas, Sigaud, & Oudeyer, 2018; Henderson et al., 2018). Our Sect. 5 contributes two items this literature does not cover for simulated robot benchmarks: the initial-state grid is finite and enumerable, so the configuration axis is a population rather than a sample and its sampling error can be driven to exactly zero; and the *implementation build* — the port branch plus the inference stack version — behaves as an unreported parameter of the same order as the ones being studied. The first changes which bootstrap is correct; the second means that even a perfectly specified bootstrap understates the uncertainty of a number copied from a published table.

3 Problem setup

3.1 Notation

Let z denote the simulator’s physical parameters (joint PD gains, torque limits, contact friction and density), and let π denote the *implementation build*: the port branch, the renderer path and the policy inference stack version. A task provides a finite set $\mathcal{C}$ of initial configurations. A policy i run from configuration c under parameters z and build π with internal random source σ either succeeds or fails, written $s_i(c, z, \sigma; \pi) \in \{0, 1\}$. The simulated difference between two policies is

$$\Delta_{ij}^{\text{sim}}(z; \pi) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} [\mathbb{E}_{\sigma} s_i(c, z, \sigma; \pi) - \mathbb{E}_{\sigma} s_j(c, z, \sigma; \pi)], \quad (1)$$

and $\Delta_{ij}^{\text{real}}$ denotes the corresponding real-robot difference. Three properties of this definition carry the weight of Sect. 5 and are stated here because they are easy to overlook: the sum over $\mathcal{C}$ is over a *population*, not a sample; σ is degenerate for deterministically decoding policies, so for those the benchmark value is measurable exactly; and $\mathcal{C}$ itself depends on π , so two builds with different grids yield two different estimands that cannot be paired per episode.

3.2 The compatible set

Calibration observes real demonstrations and scores a candidate z by a replay loss $L(z)$. Rather than reporting the minimiser $\hat{z}$, we invert a one-sided test at level α : a parameter setting is *compatible* if its per-demonstration error is not significantly worse than the best,

$$C_{\alpha} = \left\{ z : \text{lo}_{1-\alpha}(\mathbb{E}[L(z) - L(\hat{z})]) \leq 0 \right\}, \quad (2)$$

where $\text{lo}_{1-\alpha}(\cdot)$ is the paired-bootstrap $(1 - \alpha)$ lower confidence bound, so C_{α} contains every parameter the calibration data cannot reject. Directions in which the loss is *exactly* invariant (Sect. 4.2) belong to C_{α} by construction and at every magnitude, which is a stronger statement than “the data are consistent with a range of values”: there is no range, the axis is unconstrained. Parameters the replay cannot observe at all — contact friction and object density never appear

in a free-space trajectory — are likewise unconstrained, and we flag them separately because their compatibility is a property of the protocol’s coverage rather than of its precision.

3.3 The decision problem

Given C_{α} , an evaluation budget and a policy pair, we must either declare an ordering or abstain. Two criteria are compared throughout:

- **point calibration**, standard practice: evaluate at $\hat{z}$ alone and declare $i \succ j$ when the interval for $\hat{\Delta}_{ij}(\hat{z})$ excludes zero;
- **the union bound**: declare only when every $z \in C_{\alpha}$ agrees in sign, using the interval $[\min_z \text{lo}(z), \max_z \text{hi}(z)]$.

A third rung, a Gaussian-process simultaneous band over the parameter axes, is introduced in Sect. 8 for the case where the decisive parameter value may lie *between* the settings we sampled; the union bound’s guarantee is of sampling-extremum type and does not cover that case.

3.4 The error ledger

Writing $[L_{ij}, U_{ij}]$ for the union-bound interval, the claim that i is genuinely better in practice requires three events to hold simultaneously: that the true parameter lies in the compatible set, $E_{\theta} = \{z^* \in C_{\alpha}\}$ with $P(E_{\theta}) \geq 1 - \alpha$; that the evaluation intervals cover the simulated differences uniformly over the set, E_{MC} with $P(E_{MC}) \geq 1 - \gamma$; and that the sim-to-real bias is bounded, $E_b = \{|\Delta_{ij}^{\text{real}} - \Delta_{ij}^{\text{sim}}(z^*)| \leq b_{ij}\}$ with $P(E_b) \geq 1 - \beta$. On the intersection every real difference lies in $[L_{ij} - b_{ij}, U_{ij} + b_{ij}]$, and a union bound gives $1 - \alpha - \beta - \gamma$ without requiring independence.

We are deliberate about which of these this paper establishes. E_{MC} is what Sects. 5–7 measure, and enumerating $\mathcal{C}$ removes one of its two components exactly. E_{θ} is what Sect. 4 characterises, and its width dominates the ledger. E_b we do **not** establish: we have no per-pair sim-to-real bias bound, and Sect. 8.4’s delimiting experiment is best understood as evidence that on at least one pair b_{ij} is large enough to swamp everything else — 0.21 in a quantity whose other terms are 0.04 to 0.15. A criterion built on E_{θ} and E_{MC} alone bounds the ambiguity it can represent, and that is the honest description of what the ladder in Sect. 8 delivers.

4 The calibration protocol is structurally blind

4.1 Setup

Calibration of the kind used by these suites fits simulator parameters by replaying recorded real demonstrations: the recorded end-effector targets are fed to the simulated controller, and the parameters are chosen to minimise the deviation between the simulated and recorded trajectories. Let z collect the joint PD stiffness k , the damping d , the torque limit $\tau_{\max}$, the execution delay, and the contact parameters (object friction and density). Write $L(z)$ for the replay loss on the demonstration set. The demonstrations are teleoperated, free-space and slow, which is the property that drives everything in this section: the protocol does not choose its own excitation.

4.2 An exact invariance, not a weak one

Two directions in z leave L exactly unchanged.

(i) Common scaling of the PD gains.

Under free-space replay the joint response to a target is, to the order that the demonstrations excite, first order with time constant d/k . Scaling $(k, d) \mapsto (ck, cd)$ leaves the time constant, hence the trajectory, unchanged. This is a *structural* invariance of the protocol, not a flat region of a noisy objective.

(ii) The torque limit.

Along demonstrated velocities the commanded torque never reaches $\tau_{\max}$, so the saturation is inactive and the trajectory is independent of its value.

The ratio direction d/k and the execution delay *are* identified, and both produce the expected bowl-shaped loss curves; the claim is specific to the two directions above.

4.3 Empirical confirmation on two independent simulator stacks

We replayed 98 BridgeData V2 demonstrations (Walke et al., 2023) on (a) the ManiSkill3 port (Tao et al., 2024) and (b) the original reference stack (ManiSkill2 (Gu et al., 2023) with

SAPIEN (Xiang et al., 2020)), the latter running headless through a purpose-built Vulkan layer (Appendix B).

All 14 iso-scale groups fall below a 1 mm tolerance; the agreement between the two stacks is sub-millimetre. The paired error increases for the ratio direction are $+0.0089 [+0.0073, +0.0105]$ and $+0.0064 [+0.0055, +0.0072]$. Figure 1 contrasts the bowl-shaped loss along the identified directions with the flat loss along the invariant ones.

4.4 The compatible set

Inverting the calibration test therefore yields a compatible set of the form

$$C_\alpha = \{d/k \approx \hat{\rho}\} \times (\text{full scale axis}) \times (\text{full torque axis}) \times (\text{full contact axes}), \quad (3)$$

writing $\hat{\rho}$ for the identified ratio estimate, i.e. the calibration data constrains one direction tightly and leaves the others completely free. Ratio settings $\times 0.25, 0.5, 2, 4$ and the delayed-execution settings are excluded (error-increment lower bounds $+0.004$ to $+0.008$); nothing excludes any value of the common scale or the torque limit. **The question of this paper is what a policy ranking computed inside C_α is worth.**

5 What a finite benchmark measures

5.1 The benchmark is a finite population of configurations

Both widely used ports map the episode index onto a fixed grid of initial states,

$$\text{pos} = \lfloor (\text{episode_id} \bmod (n_{xy} n_{quat})) / n_{quat} \rfloor, \quad (4)$$

$$\text{quat} = \text{episode_id} \bmod n_{quat}, \quad (5)$$

so the number of distinct initial configurations is $n_{xy} \cdot n_{quat}$, and any episode index beyond that repeats an earlier scene exactly. Object scale is drawn from the asset table, but every object has a single admissible scale, so it adds no variation.

Three consequences follow, and all three are visible in published practice.

Table 1 Replay identifiability on two independent simulator stacks. 98 BridgeData V2 demonstrations. The two invariant directions are invariant to sub-millimetre and to floating-point zero respectively; the identified directions produce the expected loss increase.

Direction	ManiSkill3	Original stack	Verdict
Common scale $c \in \{0.25, 0.5, 2, 4\}$	max traj. diff. 0.141 mm	0.207 mm	invariant
Torque limit $\times 0.5$	diff. exactly 0 (bitwise)	exactly 0	invisible
Ratio $d/k \times \{0.5, 2\}$	$+0.0089$ / $+0.0064$	same sign and order	identified
Execution delay 1 step	clearly identified	clearly identified	identified

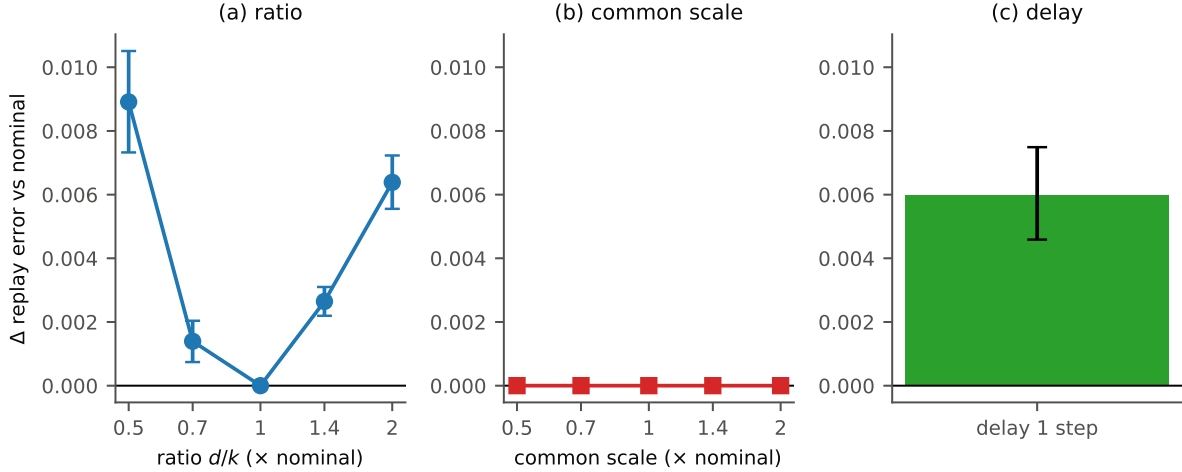


Fig. 1 Replay identifiability. Loss along the identified directions (the gain ratio d/k and the execution delay) is bowl-shaped with a well-defined minimum; loss along the invariant directions (common gain scaling, torque limit) is flat — exactly flat for the torque limit, to floating-point zero. Panels share the vertical axis.

1. **Beyond the grid, episodes are exact repeats.** A “96-episode” evaluation of the eggplant task on the ManiSkill3 port is 64 distinct scenes, 32 of which are visited twice.
2. **The two ports do not evaluate the same benchmark.** For the eggplant task they differ in the number of orientations (3 versus 8) and in the quaternion values themselves: the best matching pair has $|\langle q_1, q_2 \rangle| = 0.54$. Per-episode agreement between the ports is therefore meaningless for that task. Spoon and carrot share their grids exactly — and their cross-port per-episode agreement is 0.71–0.96, against chance level for eggplant. That contrast is the control for this diagnosis.
3. **One scene is worth 1/24 or 1/64.** On a 24-configuration task, any reported difference below 0.042 is less than one scene.

5.2 The official protocol is already a census

The reference evaluation script runs each bridge task over episode range 0 to 24 under three fixed policy seeds: the **complete 24-configuration census repeated under three fixed policy seeds**, 72 episodes, which is exactly the denominator of the published table (for instance $31/72 = 0.431$ and $41/72 = 0.569$ for the two Octo variants on the eggplant task). The ManiSkill3 port’s evaluation script instead defaults to 100 episodes indexed as seed $+i$, aligning with neither the grid nor the seed repetition. The community’s own reference protocol already does the right thing; the drift is in what downstream work does with it.

Table 2 Distinct initial configurations per task in the two ports. The eggplant task differs in both the number of orientations and the quaternion values, so an episode index does not denote the same scene across ports.

Task	Original stack	ManiSkill3 port
Put eggplant in basket	$8 \times 3 = \mathbf{24}$	$8 \times 8 = \mathbf{64}$
Put spoon on towel	$12 \times 2 = 24$	$12 \times 2 = 24$
Put carrot on plate	$12 \times 2 = 24$	$12 \times 2 = 24$
Stack green on yellow cube	$24 \times 1 = 24$	$24 \times 1 = 24$

5.3 What counts as a replicate depends on the policy

A policy seed is a replication axis only if the policy consumes it. Octo’s diffusion action head draws a pseudorandom key that enters sampling, so changing the seed produces a new sample and configuration $\times$ seed is the legitimate replication axis. OpenVLA decodes greedily with sampling disabled, so the seed is unused and changing it produces nothing new; configuration is the only axis. Empirically, two “seed sets” of the deterministic policy are record-identical in 48 of 48 episodes. For such a policy, “replicated across seed sets” is vacuous, and the configuration grid — which can be enumerated — is the only sampling axis. This is the error behind the second retraction in Sect. 1.2.

5.4 Two estimands, two intervals

Two different quantities are both legitimate, and they carry different uncertainty.

(i) *The benchmark value* — the number this benchmark reports. After a census, configuration sampling error is zero and only policy noise and run-to-run numerical noise remain, so

$$\text{Var}(\hat{\Delta}) = N^{-2} \sum_c [s_A^2(c)/S_c + s_B^2(c)/R_c], \quad (6)$$

with S_c and R_c the number of runs of each policy at configuration c .

(ii) *The generalisation value* — performance on comparable scenes. Here configuration sampling error is real and the appropriate estimator is a paired bootstrap *clustered by configuration*.

Three implementation points matter and are easy to get wrong. With a single run per configuration, estimator (i) degenerates to the ordinary binomial interval: **a census does not narrow an interval; replicates do**. Enumerate the grid first, which removes double counting, then add runs, which removes policy noise. Record-identical

re-runs of a deterministic policy must be merged into one observation, or the variance is halved for free. And estimator (ii) must cluster by configuration; in our data this widens intervals by a median factor of 1.04 on the 64-configuration task and up to 1.33 on a 24-configuration task run for 96 episodes, and changes no verdict — but the effective sample size must be reported honestly.

5.5 The evaluation protocol and the implementation build are themselves invisible parameters

Holding the configuration grid fixed, we separated three further sources.

Remark 1 (Scope of the quantities in this paper). Unless a number is explicitly labelled otherwise, every effect size and interval we report is census-scope at five seed sets: the task’s configuration grid is fully enumerated, so configuration sampling error is exactly zero, and the policy-noise term is estimated across five independent seed sets. Where a quantity is also available at the three-seed budget the reference protocol uses, we give both and say which is which, because the difference between them is itself one of the paper’s results (Sect. 7.2).

The third row deserves emphasis. `urdf_version` selects among four recolourings of the robot model. It changes no physical property — not a mass, not a friction coefficient, not a joint limit — and the published figure for this task is the mean over all four. On the fractal coke-can census the four variants give one policy 0.000, 0.280, 0.013 and 0.400, so evaluating on a single variant displaces the reported number by -0.173 to $+0.227$ relative to the four-variant mean. For comparison, the can’s physical orientation, which does change the dynamics, spans only 0.140.

This axis is invisible to replay calibration for the same reason the torque limit is (Sect. 4.2): it never enters the demonstration trajectory. Unlike

Table 3 Sources of uncertainty that persist after the configuration grid is enumerated. Policy-seed noise and the texture variant are census-scope at $S = 5$ (Remark 1). The third row is the largest in this paper’s budget and is the one a reader of the published tables cannot see.

Source	Isolation	Magnitude
Policy seed	same build, different seeds	$\text{sd}(\Delta) = 0.067$, median over pairs
Implementation build	same seeds, newer build	+0.078 paired; 0.73–0.88 of episodes reproduce
Robot texture variant	<code>urdf_version</code> only	rates 0.000 / 0.280 / 0.013 / 0.400; range 0.400

the torque limit it is not even a physical parameter, which makes it the paper’s sharpest illustration that “calibration-invisible” is a property of the protocol’s coverage rather than of physics. Whether it can flip a *ranking* rather than inflate one policy’s variance is a separate question, and Sect. 8.4 answers it on a second policy over the same grid: it does not, on the pair we can test.

Both the seed and build rows are computed on matched episode scopes. That is not a formality: the directories differ in size — the pre-fix build and one seed set hold 96 episodes against 64 elsewhere — and because episode ids beyond the grid wrap onto it, averaging each directory over its own contents compares different effective scopes. Done that way the drift reads 0.055 and would sit *below* policy-seed noise; paired on the shared ids it is 0.078 and sits *above* it. The ordering in Table 3 depends on getting this right, which is an instance of the paper’s own thesis applied to its own numbers.

Two further practical consequences follow. First, one seed set plus a per-episode bootstrap declares rankings that do not hold: in our own data $\Delta(\text{small} - \text{base}) = +0.250 [+0.08, +0.32]$ declared one Octo variant better, and three seed sets from a single build give $+0.141 / -0.031 / -0.047$. We retract that verdict. Second, replicates must never be pooled across builds; doing so inflated the measured seed standard deviation from 0.055 to 0.093, so about half of the apparent seed noise was build drift.

5.6 Reporting requirements

Every number in this paper carries: configurations covered / total; independent runs per configuration, and whether they are genuinely independent for that policy; the variance model (empirical, binomial fallback, or unmeasured); which estimand it targets; and the port and build it was collected on. We log the last of these per run in an

append-only record, having discovered that a per-directory metadata file is overwritten by the next run and makes the provenance of already-collected episodes unrecoverable.

6 Reproducing the published protocol

6.1 The protocol, followed exactly

The reference implementation’s bridge evaluation script evaluates each task over episode range 0 to 24 with three fixed policy seeds: a complete enumeration of the task’s 24 initial configurations, repeated under three fixed seeds, for the 72 episodes that form the denominator of the published rates. Within one sweep the seed is held fixed — the inference wrapper draws the same key at every reset — so the three sweeps are three draws of policy noise over one fixed configuration set, not 72 independent samples.

We reproduced this on the original reference stack, running headless through the Vulkan compatibility layer of Appendix B, with the policies served by our own inference server rather than in-process. Published values are read from the pinned metrics table distributed with the benchmark.

6.2 The reproduction agrees

For comparison, the policy-seed noise we measure on the 64-configuration eggplant census is ± 0.132 at half-width for a single seed set and ± 0.076 for three (Sect. 5.5). Every discrepancy is well inside that band. A different operating system, a software-rendered rasteriser, a different graphics path and an out-of-process policy server move the published numbers by less than one seed’s worth of noise, which is the external-validity check the rest of the paper rests on. On the second embodiment the same check gives +0.003 (octo-base on

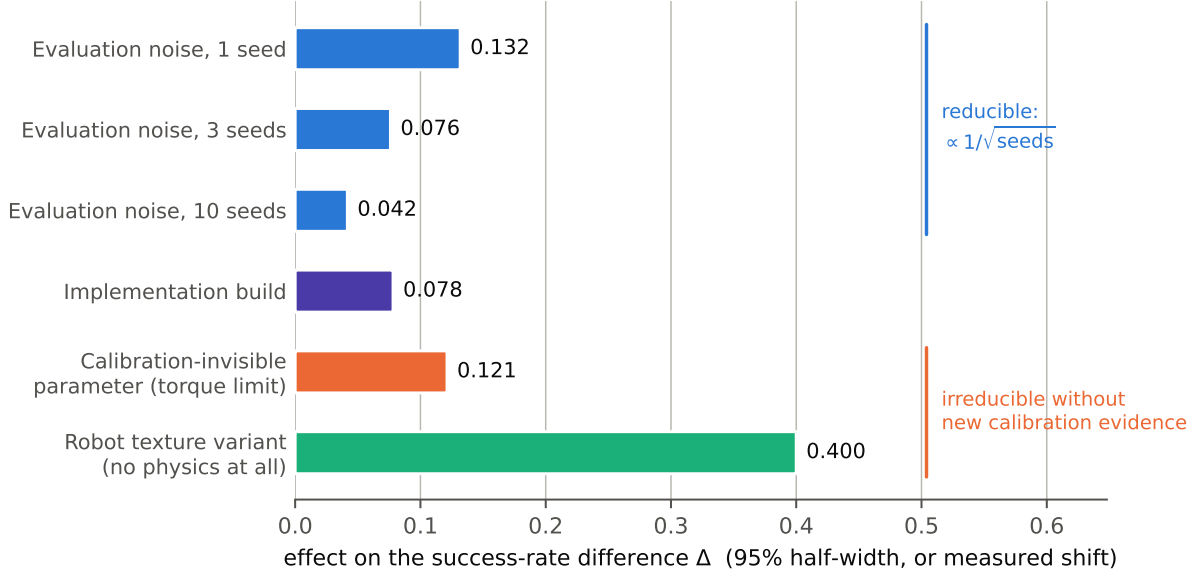


Fig. 2 The uncertainty budget after the configuration grid is enumerated. Evaluation noise at 1, 3 and 10 seeds shrinks as $1/\sqrt{S}$; the implementation build, the calibration-invisible torque limit and the robot texture variant do not shrink at all, and the largest of them is not a physical parameter. Every value is computed from the episode records.

Table 4 Our 72-episode reproduction against the published values. Across the eight cells the mean signed difference is -0.005 , the mean absolute difference 0.026 , and the largest 0.042 .

Task	Policy	seed 0	seed 2	seed 4	pooled	published	diff.
eggplant	octo-small	0.583	0.458	0.625	0.556	0.569	-0.013
eggplant	octo-base	0.375	0.458	0.583	0.472	0.431	$+0.041$
spoon	octo-small	0.375	0.333	0.583	0.431	0.472	-0.041
spoon	octo-base	0.042	0.083	0.125	0.083	0.125	-0.042
carrot	octo-small	0.042	0.042	0.125	0.069	0.097	-0.028
carrot	octo-base	0.125	0.125	0.125	0.125	0.083	$+0.042$
stack cube	octo-small	0.083	0.042	0.000	0.042	0.042	-0.000
stack cube	octo-base	0.000	0.000	0.000	0.000	0.000	$+0.000$

pick-coke-can, 0.173 against 0.170 published) and -0.040 (rt-1-x, 0.527 against 0.567).

6.3 What the published budget resolves

The same table answers a second question, and the answer is less comfortable. The three fixed seeds disagree with each other by considerably more than we differ from the published value: the across-seed range of a cell averages 0.109 and reaches 0.250 (spoon, octo-small: 0.333 to 0.583). Applying the census estimator of Sect. 5.4 to

the comparison the published protocol is built to support gives Table 5.

Three observations.

(i) One of four orderings is resolvable.

Only the spoon comparison excludes zero. On the other three tasks the published protocol reports a rate difference its own budget cannot distinguish from noise. This is not a criticism of the reference implementation — the published tables report rates, not rankings — but readers do compare those rates across policies, and on three of four tasks that comparison is unsupported.

Table 5 What the published 72-episode budget resolves. Census interval: policy noise only, $S = 3$ seeds over a complete 24-configuration enumeration. Naive interval: the binomial-over-72 a reader of the published table would compute.

Task	our Δ	census 95%	verdict at this budget	published Δ
eggplant	+0.083	$[-0.047, +0.214]$	abstain	+0.138
spoon	+0.347	$[+0.235, +0.459]$	octo-small $\succ$ octo-base	+0.347
carrot	-0.056	$[-0.137, +0.026]$	abstain	+0.014
stack	+0.042	$[-0.005, +0.089]$	abstain	+0.042

(ii) One ordering reverses.

On carrot the published values place octo-small above octo-base (+0.014); our faithful reproduction places octo-base above octo-small (-0.056). Both magnitudes are far inside the seed noise, and the published value lies comfortably within our interval, so this is not a contradiction — it is the same non-resolvable comparison resolved differently by chance in two runs of the identical protocol. It is, however, a concrete instance of the failure mode this paper is about: a ranking read off a benchmark table that a re-run of the same benchmark does not reproduce.

(iii) Removing configuration sampling error is not enough.

The census interval is tighter than the naive binomial one, but only slightly, and it changes no verdict. With three seeds per configuration the per-configuration policy variance dominates, so enumerating the configuration grid — which sets the sampling error to exactly zero — buys much less than practitioners expect. Budget is spent better on seeds than on episodes.

6.4 What this section licenses

The published numbers carry an unreported uncertainty of roughly ± 0.08 at three seeds, of the same order as the parameter-induced shift of 0.121 that Sect. 7 attributes to a calibration-invisible torque limit. That these two are comparable is the paper’s central quantitative claim, and Sect. 6.2 establishes that we can say it about the benchmark as published, not merely about our port of it.

7 Decision-relevant non-identifiability

Sections 4 and 5 showed that the calibration protocol cannot see certain parameter directions and that the benchmark’s own budget leaves its rankings unresolved. Neither observation is interesting on its own: an unidentifiable parameter matters only if moving it inside the compatible set changes an answer someone acts on. This section establishes that it can, measures how large the effect is, and reports how much of it survives when the evaluation budget is increased — which is less than we first claimed.

7.1 The claim, stated so it can fail

Fix a task, a policy pair (i, j) and the compatible set C_α of Sect. 3.2. The claim under test is that point calibration and the union bound disagree — that point calibration declares orderings the calibration data cannot support. Two forms are worth separating.

The **strong form**: some $z \in C_\alpha$ reverses a ranking’s sign, so that the simulator’s answer to which policy is better depends on a parameter the calibration data is silent about. **We report this as false.** Across every pair, task and condition we measured, no calibration-invisible parameter reverses a sign-determined ranking. We also once believed we had found a weaker version of it — a between-condition sign change on one pair — and withdrew that too (Sect. 7.5).

The **weak form**: the parameter moves Δ far enough to cross the decision boundary, turning a declared ordering into an abstention. This holds, and it is budget-dependent — which is the finding that replaced our original, stronger claim.

7.2 The core table, and what a larger budget did to it

Table 6 reports both verdicts for every policy pair on every task, with the provenance each number needs: configurations covered, independent runs per policy, and the variance model. Of 17 rows, **one** disagrees: **octo-base** vs **octo-base@hist1** on the eggplant census, point $\Delta = +0.069 [+0.00, +0.13]$ declaring **octo-base** better, union interval $[-0.06, +0.12]$ abstaining.

At three seed sets, three rows disagreed. The two that no longer do were resolved by spending seeds, not by any change of method.

The decisive cell is **octo-base@hist1** vs OpenVLA under torque $\times 0.5$, where $\Delta = +0.068$ with interval $[-0.000, +0.135]$ at $S = 3$ and $+0.073$ with $[+0.017, +0.128]$ at $S = 5$. A lower bound sitting exactly on zero became a lower bound above it, and the disagreement evaporated. We had pre-committed to this test and to withdrawing the row if it went this way, which is why the surviving count is one rather than three.

The honest general statement is therefore about the evaluation budget: at the budgets the literature actually uses — a single seed, or the reference protocol’s three — point calibration declares orderings the union bound does not support, on 3 of 17 policy pairs; at five seed sets over a complete configuration census, one remains. The criterion’s value is concentrated in the budget regime practitioners are actually in. This is weaker than the claim we set out to make, and it remains useful: we are aware of no published evaluation that runs five seed sets over a full configuration census, and the reference protocol itself uses three.

The surviving pair is also instructive about *why* it survives. It is not that its Δ changes sign across conditions (Sect. 7.5), but that its nominal interval is barely decidable (lower bound $+0.004$) while the union bound takes the minimum lower bound over conditions, which friction $\times 0.4$ supplies at -0.063 . It is a margin-and-width phenomenon, not a parameter-sign phenomenon.

7.3 The mechanism: one parameter, measured on a complete census

The torque limit is the cleanest case. It is *exactly* invisible to calibration (Sect. 4.2(ii)): along demonstrated velocities the commanded

torque never saturates, so replay loss is independent of it to floating-point zero. Halving it therefore stays inside C_α by construction, not by a fitting tolerance. On the 64-configuration eggplant census it moves the cross-family comparison by -0.104 , -0.146 , -0.108 and -0.127 for the four Octo-versus-OpenVLA pairs: a shift of 0.104 – 0.146 , mean **0.121**, uniform enough across pairs to be consistent with a single policy moving rather than four independent effects (`scripts/analyze_torque_shift_s5.py`).

Remark 2. Two earlier measurements of this shift were larger, and both were superseded for the same kind of reason. On 48 of the 64 configurations it read 0.23 – 0.25 , with two pairs crossing the decision boundary: the partial census overstated it. On the complete census at three seed sets it read 0.133 – 0.161 , mean 0.141 ; at five it reads as above. Every effect size in this paper is census-scope at $S = 5$ for this reason, which is the budget Sect. 7.2 establishes as the one that settles verdicts.

7.4 Why it is not an artefact

Three checks.

It is a property of one policy, and a large one.

Halving the torque limit raises OpenVLA’s eggplant benchmark value from 0.156 to 0.271 on the complete 64-configuration census ($+0.115$), while octo-small, run as a negative control over the same census, moves by $+0.011$ and the other three Octo policies by -0.06 to $+0.02$ with no consistent direction. The asymmetry has a mechanism: OpenVLA emits larger single-step displacements, so at the nominal limit it pushes the eggplant away or re-grasps repeatedly on contact, and the episode-level counters agree (grasp rate $+0.12/+0.18$, fewer repeated gripper closures).

It is broad, not driven by a few configurations.

Over the complete census 16 configurations improve, 42 are unchanged and 6 degrade, and the mean change is positive in 7 of the 8 eggplant orientations (-0.04 to $+0.29$). The gain is anticorrelated with nominal success ($r = -0.44$): of the 51 configurations where OpenVLA fails outright

Table 6 The effect of raising the seed budget from three to five on the point-versus-set disagreements. Two of the three were our own insufficient power.

Pair	Union at $S = 3$	at $S = 5$	Outcome
octo-small vs octo-base@hist1	$[-0.03, +0.28]$ abstain	$[+0.01, +0.24]$ declare	withdrawn
octo-base@hist1 vs OpenVLA	$[-0.00, +0.42]$ abstain	$[+0.02, +0.43]$ declare	withdrawn
octo-base vs octo-base@hist1	$[-0.10, +0.17]$ abstain	$[-0.06, +0.12]$ abstain	survives

at the nominal limit, 13 are rescued. The parameter repairs failures rather than padding successes. The same decomposition for octo-small is flat and directionless — 15 improve, 15 degrade, 2 of 8 orientations positive — which is what a policy that never saturates the limit should look like.

It is task-dependent in the direction the mechanism predicts.

On the spoon task (24-configuration complete census) the same parameter barely moves Δ at all — -0.01 to -0.07 across the four cross-family pairs, against -0.14 on eggplant — and changes no verdict. Eggplant is where contact torque saturates for a large-displacement policy; spoon is not. A calibration-invisible parameter is therefore not uniformly dangerous: its decision-relevance is a property of the task-policy pair, which is precisely why it cannot be argued away a priori and has to be enumerated.

7.5 A withdrawn claim: there is no between-condition sign reversal

An earlier version of this section reported that one pair, **octo-base** vs **octo-base@hist1**, reversed sign across the compatible set — $\Delta = -0.021$ under density $\times 0.5$ against positive values under the other five conditions — and argued from it that this pair’s abstention was a fact about the parameters rather than about our budget, and so could never be removed by more evaluation. **We withdraw that.** At five seed sets the same cell reads $+0.042 [-0.022, +0.106]$, and all six conditions have point estimates at or above -0.000 : nominal $+0.069$, force $\times 0.5$ $+0.050$, iso $\times 0.25$ $+0.030$, iso $\times 4.0$ $+0.006$, friction $\times 0.4$ -0.000 , density $\times 0.5$ $+0.042$.

The distinction the withdrawn claim was reaching for is nevertheless real and worth keeping, because Sect. 8 needs it. As the evaluation half-width $h \rightarrow 0$ the union bound converges to $[\min_c \Delta_c, \max_c \Delta_c]$, which does **not** shrink with

budget. An abstention that survives $h \rightarrow 0$ is a parameter fact; one that dissolves as h falls was a power artefact. Our own three-to-five-seed experiment is the clean demonstration of the second case, and Sect. 8.4 uses the same test to reject an abstention we would otherwise have been tempted to claim as a success.

7.6 What determines whether a ranking can be declared

A ranking can be declared when $|\hat{\Delta}|$ exceeds the parameter-induced shift δ **plus** the evaluation half-width h . Enumerating the configuration grid drives the sampling part of h to zero; more seeds shrink the rest, as $1/\sqrt{S}$ — from 0.075 – 0.088 at three seed sets to 0.058 – 0.068 at five. The parameter-induced shift $\delta \approx 0.121$ does not move, by construction.

The two regimes this produces are the practical content of the section. At the budget the reference protocol uses — 24 configurations, three seeds — a pair’s half-width is 0.05 – 0.13 (Sect. 6.3), which is at or above δ ; there the two error sources are not even separable, and a practitioner cannot tell an identifiability problem from a power problem. Our census at $S = 5$ pushes h below δ , and only then does the comparison become meaningful. The recommendation follows directly: report the estimand and the budget, and spend budget on seeds before episodes, because only then does the residual ambiguity that *cannot* be bought down become visible.

8 A verdict ladder and its reach

Sections 4 through 7 establish that a simulation benchmark’s ranking claims are underdetermined by the evidence behind them, from three independent directions. This section asks what to do about it: what a set-valued criterion buys, at what cost

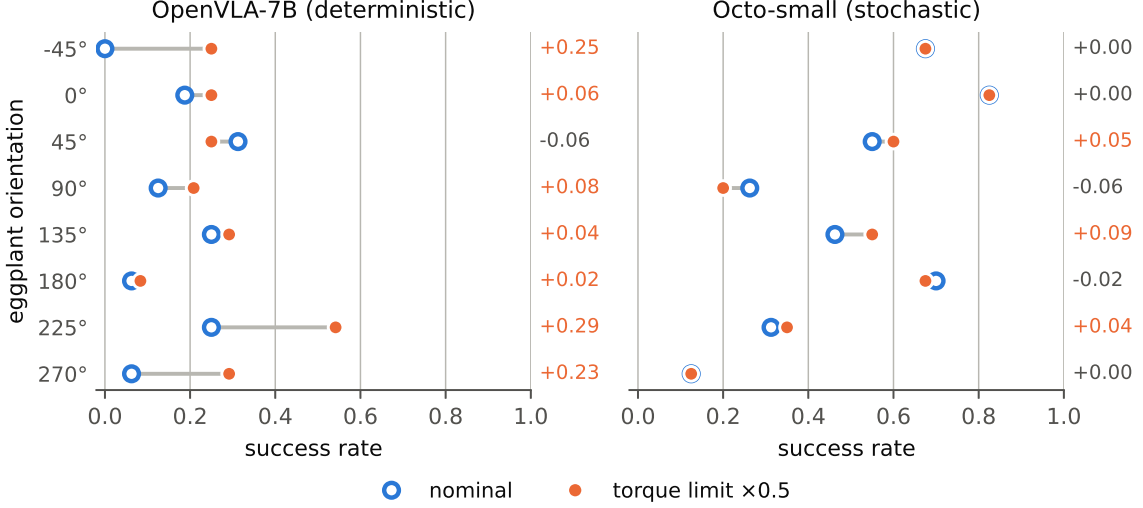


Fig. 3 The torque-limit effect decomposed over the eggplant orientation grid, for a deterministically decoding policy (OpenVLA, left) and a stochastic one (Octo-small, right). The effect is positive in 7 of 8 orientations for the former and directionless for the latter, which is the asymmetry the saturation mechanism predicts.

in power, and where it stops working. We answer all three, and the third answer is negative.

8.1 Two rungs, with different guarantees

The **union bound** declares an ordering when every *sampled* parameter setting in C_α agrees in sign. Its guarantee is of **sampling-extremum** type: valid with respect to the extremes of $\Delta(z)$ over the settings actually run, and silent about values between them.

A **Gaussian-process simultaneous band** fits $\Delta(z)$ over the parameter axes and requires the sign to hold across a simultaneous band over the whole compatible region. Its guarantee is of **within-band** type under a smoothness prior, and it therefore covers features that fall between sampled settings — at a cost in power that Sect. 8.2 quantifies.

An earlier design had a third element, a flatness gate that chose between the two per axis. We removed it; Sect. 8.2 explains why.

8.2 Coverage and power on synthetic ground truth

Because real data provide no ground truth about $\Delta(z)$, we built a synthetic track in which the

response surface is known and the criteria can be scored directly: 300 repetitions per cell over four surface families, crossed with true margins Δ_0 and episode budgets n .

(i) Point calibration’s error rate rises with the episode budget.

On any non-flat surface the nominal interval covers $\Delta(\hat{z})$ rather than the extremes over C_α , so more episodes shrink the interval around a value that was never the quantity of interest. On the linear surface at $\Delta_0 = 0.1$ the error rate goes from 0.13 at $n = 24$ to 0.36 at $n = 96$; on the sampled-dip surface at $\Delta_0 = 0.2$, from 0.26 to 0.88. This is the quantitative form of “more evaluation makes a wrong conclusion more confident”.

(ii) The union bound attains its guarantee, and only its guarantee.

Where the extremum of $\Delta(z)$ falls on a sampled setting it has coverage ≥ 0.98 with error rate 0 and the highest power of any valid rung (declaration rate 0.98 on the flat surface at $\Delta_0 = 0.3$, $n = 96$). Where the decisive feature falls between sampled settings it fails outright: coverage 0.22, error rate 0.17.

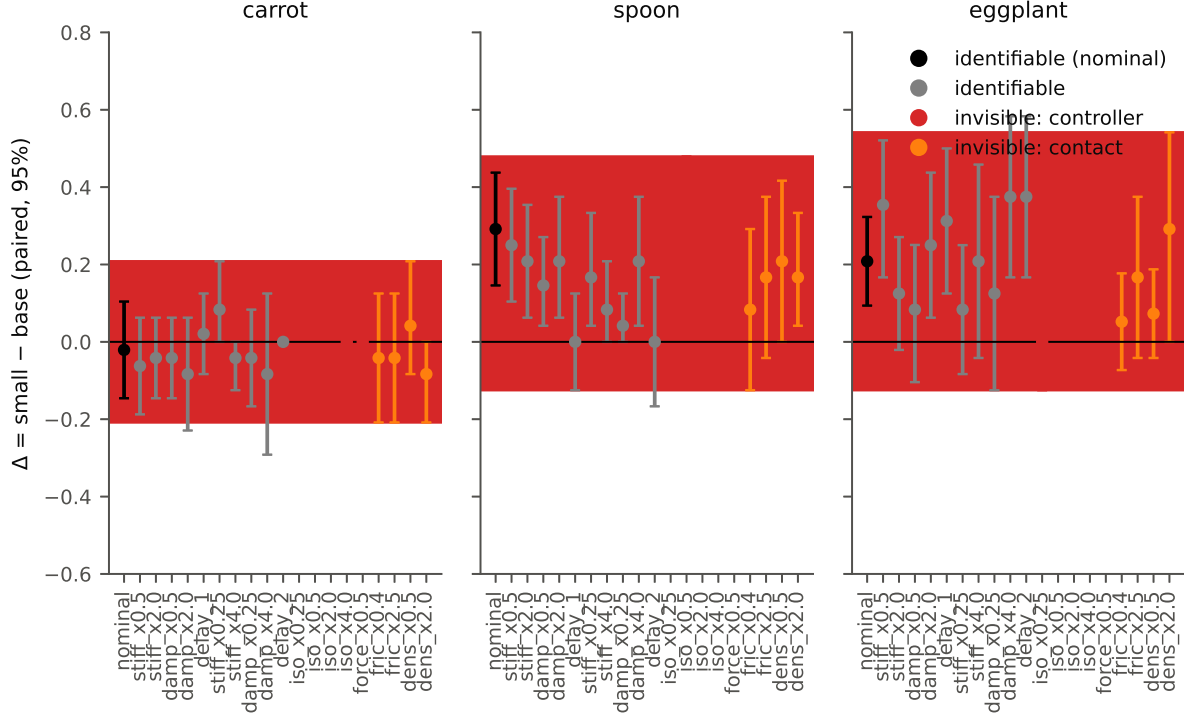


Fig. 4 Δ between two Octo variants by simulator condition, on three bridge tasks. Grey points are conditions the calibration data identifies, coloured points are the calibration-invisible ones, and the shaded band is the union bound over the invisible conditions. Verdicts and set bounds over the invisible conditions: carrot, point abstains and set abstains, $[-0.21, +0.21]$; spoon, point declares octo-small better but the set abstains, $[-0.12, +0.48]$; eggplant, point declares octo-small better but the set abstains, $[-0.12, +0.54]$. Episodes per condition range from 24 to 96 and are tabulated in Appendix C.

Table 7 Minimum coverage / maximum error rate / mean declaration rate on cells where the truth is decidable, by response-surface family. The union bound fails exactly where the decisive feature falls between sampled settings.

Surface	Point calibration	Union bound	GP simultaneous band
Flat	0.92 / 0.05 / 0.53	1.00 / 0.00 / 0.24	1.00 / 0.00 / 0.14
Linear	0.24 / 0.36 / 0.71	0.99 / 0.00 / 0.29	0.95 / 0.00 / 0.15
Dip at a sampled setting	0.00 / 0.88 / 0.45	0.98 / 0.00 / 0.03	0.97 / 0.00 / 0.04
Dip between settings	0.00 / 0.85 / 0.43	0.22 / 0.17 / 0.21	0.91 / 0.00 / 0.08

(iii) *The GP band covers all four families at ≥ 0.91 with error rate 0, and pays for it in power.*

Its declaration rate on the flat surface at $\Delta_0 = 0.3$, $n = 96$ is 0.47, and barely grows from $n = 48$ to $n = 96$, because the band extrapolates half a grid step beyond the sampled range and the posterior diverges at the ends. Restricting the band to the sampled range raises that rate to 0.77 and restores monotonicity in n , at a coverage cost on

the between-settings dip (0.91 to 0.85). We adopt the restricted configuration.

We also report a configuration we **rejected**, because it is a methodological point rather than a tuning detail. Imposing a lower bound on the GP length scale equal to the sampling spacing doubles power again (flat-surface declaration 0.22 to 0.40, linear 0.17 to 0.30) but assumes the surface is smooth at exactly the scale where the dip families are not, and coverage collapses to 0.45–0.83 with error rates up to 0.10. A response-surface

method’s power can be bought only with prior assumptions, and the stronger the prior the more brittle the method is to the features it did not sample. The flatness gate we removed is the same lesson: with 13 design points the GP already fits the sampled-dip surface, so the gate only lowered coverage on the between-settings surface (0.91–0.98 to 0.89) and mis-fired on flat surfaces 16–20% of the time.

8.3 Stability on real replicate data

Synthetic ground truth cannot tell us whether the real surfaces resemble any of the four families. So we also score the criteria on real data by replication: a verdict is *stable* if an independent replicate of the same evaluation reproduces it.

Point calibration reproduces half of its own verdicts; the union bound over the controller axes reproduces 0.88 of them. But the last row must be read with its final column: the all-axes union bound achieves perfect stability by **never declaring anything** on this data. That is not a success, it is the degenerate end of the conservatism dial — a criterion that always abstains is perfectly reproducible and perfectly useless. The useful reading is the middle row, where a real gain in stability (0.50 to 0.88) is bought with a real loss in declarations (0.38 to 0.06).

8.4 The delimiting experiment: a pair the real robot reverses

Everything to this point is internal to the simulator. The criterion is conservative where we can check conservatism and stabilises verdicts where we can check stability, but neither test asks whether it protects against being *wrong about the real robot*. There is one place in this benchmark where that can be asked without hardware.

The reference implementation publishes real-robot and simulated success rates for the same policies on the same task. On pick-coke-can the two orderings disagree on three of fifteen policy pairs, all with small real margins; of those, exactly one has both checkpoints publicly available (Brohan et al., 2023).

The question is whether the union bound *abstains* on a pair the simulator gets wrong. It is the sharpest available test of the criterion, because we know the answer the simulator should not have

given. We ran both checkpoints over the complete 300-configuration census under nominal conditions plus the two calibration-invisible conditions that moved verdicts on bridge — the torque limit (exactly invisible, Sect. 4.2) and object friction — for 1,800 episodes total. The result is adverse. Point calibration on the nominal condition gives $[+0.083, +0.210]$, declaring **rt-1-converged** better. The union bound over the dynamics compatible set gives $L = +0.070$, $U = +0.240$ — **also declaring rt-1-converged better**, which is the ordering the real robot reverses.

Widening from the point to the set moved the lower bound from $+0.083$ to $+0.070$. The three conditions span margins of 0.133 to 0.177 — a total spread of 0.044 — while the correction needed to reach abstention is about 0.21. The torque limit and object friction are nowhere near the binding constraint on this pair.

Nor does the largest hidden variable we know of rescue it. Because this run puts a second policy on the same 300-configuration grid, it also settles the question left open by Sect. 5.5: whether a visual variant that carries no physics can flip a *ranking* rather than merely inflate a single policy’s variance. Adding `urdf.version` as a fourth set dimension (12 cells of 75 configurations each) does turn the verdict to abstention, $L = -0.067$ — but the abstention is of the second kind identified in Sect. 7.5. All twelve cells have **positive** point margins, from $+0.053$ to $+0.213$, and not one places **rt-1-15pct** above **rt-1-converged** with interval support. The bound reaches zero only through interval width at 75 configurations per cell, so as the budget grows L tends to $+0.053$ and the abstention dissolves. By our own test it is our power, not a parameter fact, and we decline to claim it as a success of the criterion.

The conclusion we draw is about the criterion’s reach. The sim-to-real gap on this pair is roughly 0.21 in the margin, and 0.20 in one policy’s own rate (**rt-1-15pct**: 0.920 real against 0.717 simulated). No direction in the dynamics compatible set, and not the texture axis either, moves the simulated margin by more than 0.044. Whatever produces the reversal lies outside the parameter family that demonstration replay can constrain — which is, in the ledger of Sect. 3.4, a statement that b_{ij} is large on this pair and that a criterion built on E_θ and E_{MC} cannot see it.

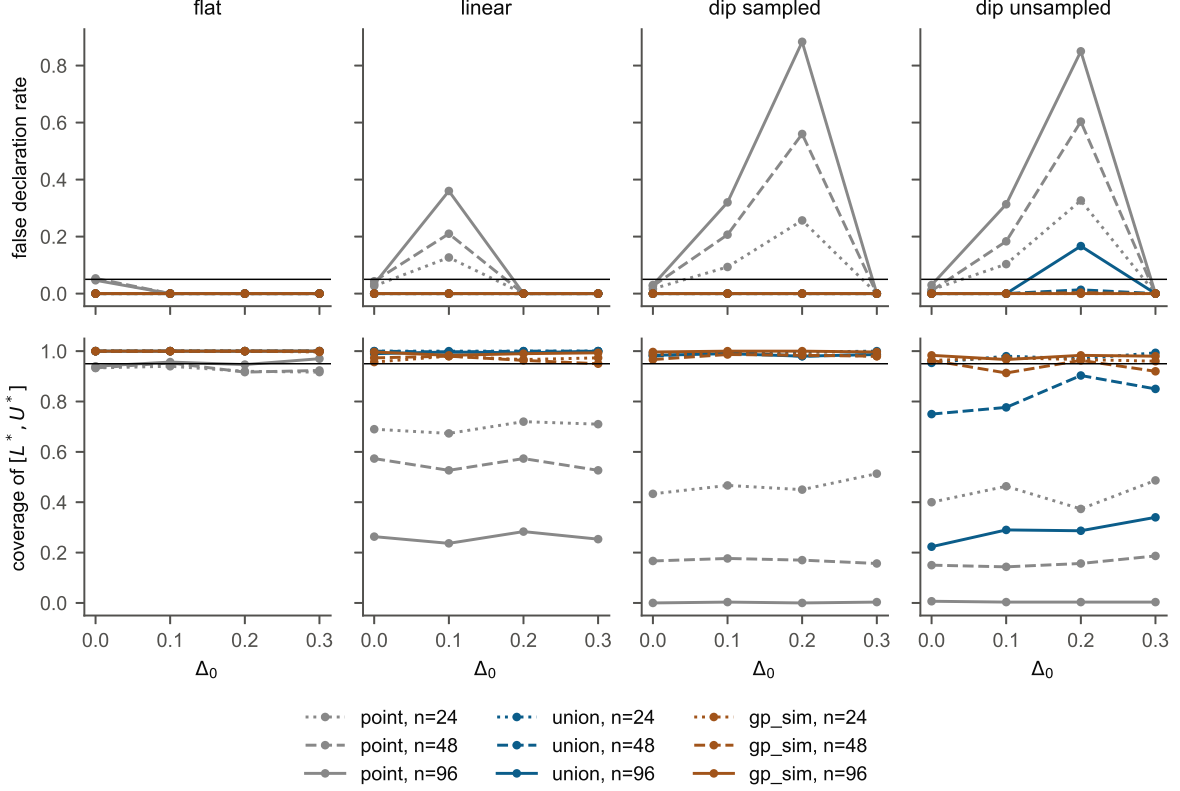


Fig. 5 The three criteria on synthetic ground truth, by response-surface family (columns), true margin Δ_0 (horizontal axis) and episode budget (line style). **Top row:** false declaration rate, with the nominal 0.05 level marked. **Bottom row:** coverage of $[L^*, U^*]$, with the nominal 0.95 level marked. Point calibration (grey) gets *worse* as the budget grows on every non-flat surface — its false declaration rate reaches 0.88 at $n = 96$ on the sampled-dip surface, and its coverage falls towards zero. The union bound (blue) holds at ≥ 0.98 coverage except on the between-settings dip, where it drops to 0.22 and declares falsely at up to 0.17; the GP simultaneous band (orange) holds on all four families. The withdrawn flatness gate is not plotted (Sect. 8.2).

Table 8 Verdict stability over the eight replications whose two sides enumerate the same configuration grid. The last row attains perfect stability by never declaring; the informative row is the middle one.

Criterion	Pairs	Stable	Contradict	Unsupported	Declare
Point calibration	8	0.50	0.00	0.50	0.38
Union bound, controller axes	8	0.88	0.00	0.12	0.06
Union bound, all axes	8	1.00	0.00	0.00	0.00

8.5 What the ladder is for

Taken together the three tests place the criterion precisely. It bounds the ambiguity the calibration data leaves in the parameters it can represent: there it is conservative by construction (Sect. 8.2), it makes verdicts reproducible where point calibration is a coin flip (Sect. 8.3), and it changes

the verdict on 3 of 17 pairs at the budgets practitioners use (Sect. 7.2). It does not bound, and cannot bound, the ambiguity arising from a model gap that no setting of those parameters expresses — and Sect. 8.4 exhibits a pair where that second ambiguity is five times the first and points the other way.

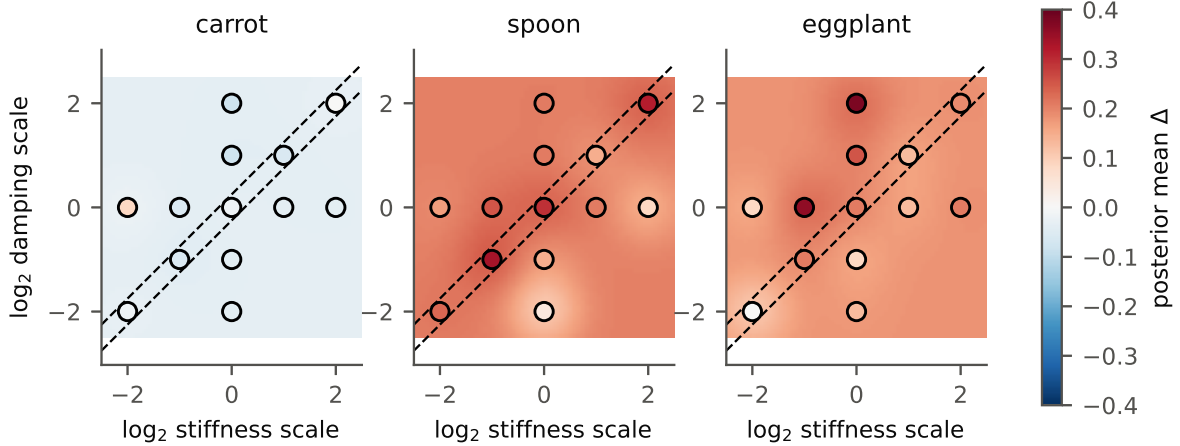


Fig. 6 Gaussian-process posterior mean of the policy difference over the controller plane, for three bridge tasks; circles are the 13 sampled design points and the dashed lines bound the replay-compatible strip (gain ratio ≈ 1). Strip bounds at $k = 2$ and the resulting verdicts: carrot $[-0.08, +0.01]$, abstain; spoon $[+0.10, +0.33]$, octo-small better; eggplant $[+0.03, +0.30]$, octo-small better. Fitted length scale $\ell = 0.5$ throughout, signal $\sigma_f = 0.02$ (carrot) and 0.05 (spoon, eggplant). The band is restricted to the sampled range, the configuration adopted in Sect. 8.2.

Table 9 The real-versus-simulation reversal pair. Our reproduction of the published simulated rates is $+0.006$ and $+0.007$, so the disagreement is reproduced faithfully rather than introduced.

Source	rt-1-converged	rt-1-15pct	Margin
Published real robot	0.853	0.920	-0.067 (15pct better)
Published simulation	0.857	0.710	$+0.147$ (converged better)
Our nominal census	0.863	0.717	$+0.147$

We therefore do not recommend the union bound as a device for deciding which policy is better. We recommend it as a device for detecting when a benchmark’s own evidence cannot decide, which is a weaker and more defensible role, and for which Sects. 8.2 and 8.3 are the relevant evidence. A declaration from it is not a sim-to-real guarantee; Sect. 8.4 is the counterexample that makes that sentence necessary.

9 Limitations

We list these in descending order of how much they should change a reader’s confidence, rather than in the order that flatters the paper.

1. **The criterion’s central negative result rests on a single policy pair.** Section 8.4 shows the union bound declaring an ordering the real robot reverses, and that is one pair, on one task, on one embodiment. It is the only

pair in this benchmark for which the published real and simulated orderings disagree *and* both checkpoints are public, so it is the only such test available to us — but a single counterexample establishes that the criterion’s reach is limited, not how often the limit is reached. A reader should conclude that a union-bound declaration is not a sim-to-real guarantee, and should not conclude that such declarations are usually wrong.

2. **No hardware of our own.** Every simulated number here is ours; every real-robot number is quoted from the reference implementation’s published tables. Those are the values the benchmark itself is validated against, and using them is what makes Sect. 8.4 possible without a laboratory, but they carry their own unreported uncertainty: they are single-protocol real evaluations, and the pair in Sect. 8.4 has a real margin of 0.067 , of the same order as the

seed noise we measure in simulation. A replication of that pair on a real robot is the single most valuable follow-up, and it is out of our reach.

3. **Narrow coverage of policies, embodiments and tasks.** Three model families (two Octo variants with three deployment configurations, OpenVLA, two RT-1 checkpoints), two embodiments and four tasks. OpenVLA is run 4-bit quantised, which we did not control against a bf16 baseline; the torque mechanism in Sect. 7.4 is a claim about the quantised model as deployed. The dependence of decision-relevance on the task-policy pair means our coverage is exactly the wrong shape for extrapolation: we show the effect exists on one task and is nearly absent on another, and we cannot say which is typical.
4. **The strong form of the non-identifiability claim is false in our data.** No calibration-invisible parameter reverses a sign-determined ranking anywhere we looked, and the one between-condition sign reversal we once claimed did not survive a larger budget (Sect. 7.5).
5. **The compatible set is sampled, not covered.** The union bound evaluates a discrete set of parameter settings, so its guarantee is of sampling-extremum type, and Sect. 8.2 shows it failing (coverage 0.22) precisely when the decisive feature lies between sampled settings. On the contact axis we have three sampled points, too few for the GP rung to be trustworthy, so those axes use the union bound and inherit its limitation.
6. **Low success rates produce many abstentions, and some are ours rather than the data’s.** Several tasks sit at success rates of 0.1 to 0.5, where even a complete 24-configuration census at three seeds gives half-widths of 0.05 to 0.13. We applied the $h \rightarrow 0$ test wherever the budget allowed, but for the original-stack rows of Table 6, which have one run per configuration, we cannot.
7. **Implementation-build drift is separated on two policies only.** The 0.078 figure and the 0.73–0.88 episode-level agreement come from a controlled re-run of two Octo policies across inference-stack versions. We treat π as an axis of the same kind as z on that

basis, which is thinner evidence than the rest of Sect. 5 rests on.

8. **Part of the platform work is exploratory.** The ManiSkill3 configuration was used for the sweeps, with the original reference stack used for cross-checking; the two agree on the replay invariances to sub-millimetre and on the union-bound verdicts in Table 6, but they differ on point verdicts for one task, and their eggplant configuration grids differ so that per-episode comparison is undefined there. Rows are labelled by stack throughout and should not be pooled.

10 Conclusion

We asked which ranking claims a calibrated simulation benchmark’s own evidence determines, and answered it on a widely used sim-to-real evaluation suite. The calibration protocol is exactly uninformative about a common scaling of the joint gains and about the torque limit — 0.141 mm and bitwise zero on two independent simulator stacks. The benchmark’s initial states form a finite population of 24 to 300 configurations rather than a sample, so enumerating them sets configuration sampling error to zero, and two widely used ports enumerate different populations. And following the published protocol exactly reproduces its rates to 0.026 while showing that its own 72-episode budget resolves one of the four orderings it reports, with a second reversing sign under re-run.

The quantitative comparison these three findings converge on is the paper’s practical message. On the eggplant census, a single seed set gives a half-width of ± 0.132 ; the reference protocol’s three seeds give ± 0.076 ; ten seeds would give ± 0.042 . Against that, the shift induced by one calibration-invisible parameter is 0.121, the spread across the four robot texture variants the benchmark silently averages over is 0.400, and the sim-to-real gap on the one pair we can check against real ground truth is 0.21 — pointing the other way. **The terms that evaluation budget can reduce are the small ones.** A practitioner who responds to an ambiguous benchmark result by running more episodes is buying down the least significant digit, and — on any non-flat response surface — is making a possibly wrong conclusion more confident while doing so.

What we recommend follows from where each term sits. Report the estimand and the budget, because a rate difference means something different as a benchmark value over an enumerated population than as a generalisation value over sampled scenes, and the correct interval differs accordingly. Enumerate the configuration grid, since it is finite and the marginal episode past it carries no information. Report which implementation build and which visual variant a number came from, because the variant axis carries the largest variance in our budget and is currently averaged over without mention. Spend budget on seeds before episodes. And use a set-valued verdict — but for the role it can actually fill.

That last point is where our own expectations were corrected. We built the compatible-set criterion expecting to show that it repairs unreliable rankings, and what we can show is narrower: it detects when a benchmark’s evidence cannot decide, it makes verdicts reproducible where point calibration is a coin flip, and it changes the answer on 3 of 17 pairs at the budgets practitioners use — falling to 1 of 17 when we spend five seed sets over a full census. It does not protect against a model gap that no value of the parameters expresses, and on the single pair where published real-robot results contradict the simulator, it declares the ordering the real robot reverses. A criterion of this kind bounds the ambiguity it can represent. On at least one pair that is not the ambiguity that matters, and we know of no way to discover that fact from simulation alone.

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Use of generative artificial intelligence.

A large language model (Anthropic Claude) was used substantially in preparing this work, and the authors consider it appropriate to describe that use rather than characterise it merely as language editing. It was used to draft and revise manuscript prose; to write analysis and figure-generation code, including the scripts that produce the tables and figures reported here; to convert the manuscript into the journal’s template; and to check bibliographic records against

publisher sources. The study design, the experiments, the interpretation of results, and every decision to assert or withdraw a claim are the authors’ own. All quantitative results are computed from the episode records by the released scripts rather than produced by a language model, and the authors have verified the content and take full responsibility for it. No language model is listed as an author, in accordance with journal policy.

Declarations

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Consent for publication. Not applicable.

Data availability. All episode-level records underlying the results are released with the code repository as append-only JSON Lines files, one per policy, task and simulator condition, together with the per-run provenance log that records the port branch, inference-stack version and random seeds for every episode. The real-robot reference values used in Sect. 8.4 are the published values distributed with the benchmark’s own source and are not ours to redistribute.

Materials availability. Not applicable.

Code availability. The evaluation harness, the configuration-census utilities, the six resumable evaluation queues, the analysis scripts that generate every table and figure in this paper, and the Vulkan compatibility layer described in Appendix B are released under an open licence.

Author contributions. **Jun Ji:** Data curation, Methodology, Formal analysis, Resources, Writing – original draft. **Bowen Tan:** Software, Formal analysis, Writing – review & editing. **Yi Li:** Supervision, Project administration, Writing – review & editing. **Xiaolei Zhang:** Supervision, Project administration, Writing – review

& editing. **Shengjie Guo**: Supervision, Project administration, Writing – review & editing. **Yi Sui**: Supervision, Validation, Writing – review & editing.

Appendix A Test inversion for the compatible set

A.1 Construction

Calibration observes M real demonstrations. For a candidate parameter z , let $L_m(z)$ be the replay error on demonstration m — the deviation between the simulated and recorded trajectories — and let $\hat{z}$ minimise $\bar{L}(z) = M^{-1} \sum_m L_m(z)$ over the grid we search.

Rather than report $\hat{z}$, we invert a one-sided test. For each candidate z we form the paired per-demonstration differences $\{L_m(z) - L_m(\hat{z})\}_{m=1}^M$, resample demonstrations with replacement B times, and take $\text{lo}_{1-\alpha}(z)$ to be the α quantile of the bootstrap distribution of their mean. Then

$$C_\alpha = \{z : \text{lo}_{1-\alpha}(z) \leq 0\}, \quad (\text{A1})$$

the set of parameters whose replay error is not significantly worse than the best. Pairing by demonstration matters: the demonstrations differ enormously in difficulty, and an unpaired test compares that between-demonstration spread against the parameter effect rather than against itself.

Three properties of this construction carry the argument of Sect. 4.

Exactly invariant directions belong to C_α at every magnitude.

If $L_m(z) = L_m(z')$ for all m , then every paired difference is identically zero, the bootstrap distribution is a point mass at zero, and $\text{lo}_{1-\alpha} = 0 \leq 0$. Membership does not depend on α , on B , or on M . This is stronger than “the data are consistent with a range of values”: along such a direction there is no range, because the axis is unconstrained at every magnitude. Increasing M does not shrink it.

Unobserved parameters are compatible for a different reason.

Contact friction and object density never enter a free-space replay trajectory at all, so their compatibility reflects the protocol’s *coverage* rather than its precision. We flag these separately throughout, because a reader might otherwise conclude that a longer or better-excited demonstration set would constrain them. It would not; a different experiment is needed.

Identified directions are excluded with margin.

Along the gain ratio and the execution delay the same procedure returns strictly positive lower bounds, +0.004 to +0.008 (Sect. 4.3), so those settings are rejected rather than retained. The construction therefore discriminates; it is not vacuously inclusive.

A.2 The error ledger, and what this paper does not establish

Writing $[L_{ij}, U_{ij}]$ for the union-bound interval of Sect. 3.3, the claim that policy i is genuinely better in practice requires three events at once:

$$E_\theta = \{z^* \in C_\alpha\}, \quad (\text{A2})$$

$$E_{MC} = \{\forall z \in C_\alpha, \forall \pi \in \Pi, \forall (i, j) : \Delta_{ij}^- \leq \Delta_{ij}^{\text{sim}}(z; \pi) \leq \Delta_{ij}^+\}, \quad (\text{A3})$$

$$E_b = \{\forall (i, j) : |\Delta_{ij}^{\text{real}} - \Delta_{ij}^{\text{sim}}(z^*)| \leq b_{ij}\}, \quad (\text{A4})$$

with $P(E_\theta) \geq 1-\alpha$, $P(E_{MC}) \geq 1-\gamma$ and $P(E_b) \geq 1-\beta$. On $E_\theta \cap E_{MC} \cap E_b$ every real difference lies in $[L_{ij} - b_{ij}, U_{ij} + b_{ij}]$ simultaneously, and a union bound gives at least $1-\alpha-\beta-\gamma$ without any independence assumption. Note that E_{MC} is stated over $\Pi \times C_\alpha$: the implementation build is treated as an axis of the same kind as the physical parameters, for the reason given in Sect. 5.5.

Each event needs its own argument, and four caveats are worth stating explicitly because each is a way the ledger can be misused.

1. **Per-pair bias coverage is not the joint event.** Establishing $|\Delta_{ij}^{\text{real}} - \Delta_{ij}^{\text{sim}}| \leq b_{ij}$ for each pair separately does not give E_b , which

is a statement over all pairs at once. A per-pair guarantee must be inflated for multiplicity before it can enter the ledger.

2. E_θ **presumes that z^* exists.** The event is empty of content if no parameter setting reproduces reality — that is, if the simulator is structurally misspecified rather than merely uncalibrated. The fitted parameters may then be no more than locally effective quantities, and α bounds nothing. Section 8.4 is best read as a case where this premise is under strain: no setting of the parameters we can vary reproduces the observed real ordering.
3. **A Gaussian-process posterior standard deviation is not γ .** Substituting the posterior spread of the response-surface rung into γ without validating its calibration would assume the very smoothness the rung is there to test. Section 8.2 measures that rung’s coverage on synthetic surfaces instead, including surfaces that violate its prior.
4. **Surrogate error must be bounded to be absorbed.** Where an approximation has a certified error bound, the band can be widened by it. Where it does not, that rung is an empirical diagnostic and must not be reported as a coverage guarantee.

Two further points concern variance reduction rather than coverage. Pairing on common initial states and common exogenous random draws reduces the variance of $\hat{\Delta}$ and we use it throughout, but it does not by itself produce a simultaneous band over C_α — simultaneity has to be constructed, not inherited from pairing. And the internal action sampling of two different policies does not correspond; forcing it into a pairing and treating the result as a shared exogenous environment is not valid, which is why Sect. 5.3 treats the seed as a replication axis only for policies that consume it.

What this paper establishes. Sections 5–7 measure E_{MC} , and enumerating $\mathcal{C}$ removes one of its two components exactly. Section 4 characterises E_θ , whose width dominates the ledger. We do **not** establish E_b : we have no per-pair sim-to-real bias bound, and Sect. 8.4 exhibits a pair on which b_{ij} is large enough to swamp every other term.

Appendix B Headless rendering of the original stack without a GPU

The published-protocol reproduction of Sect. 6 and the fractal experiment of Sect. 8.4 both require the *original* reference stack rather than the ManiSkill3 port, because the two enumerate different configuration grids (Sect. 5.1). That stack renders through SAPIEN’s `svulkan2`, which requests `VK_KHR_external_semaphore_fd` at device creation. Under WSL2 on Mesa lavapipe no driver offers file-descriptor-based external semaphores, so device creation fails and the renderer never initialises — on a host with no GPU there is no path that satisfies the request.

The extension is requested but, for our workload, unused: fd-based external semaphores serve cross-API synchronisation, which `svulkan2` exercises only when CUDA interop is enabled. We therefore supply an explicit Vulkan layer, `VK_LAYER_the_world_fakesemfd`, that makes the request satisfiable without providing the underlying capability. In 210 lines of C it:

1. advertises `VK_KHR_external_semaphore_fd` and `VK_KHR_external_fence_fd` in `vkEnumerateDeviceExtensionProperties`, appending them to whatever the driver reports;
2. strips those two names from `ppEnabledExtensionNames` at `vkCreateDevice`, so the driver is never asked to honour them;
3. removes `VkExportSemaphoreCreateInfo` and `VkExportFenceCreateInfo` from create-info chains; and
4. stubs the fd import and export entry points, which are consequently never reached.

It is an *explicit* layer, enabled per process through `VK_LAYER_PATH` and `VK_INSTANCE_LAYERS` and installed nowhere:

```
export VK_LAYER_PATH=<dir>
export VK_INSTANCE_LAYERS=\
    VK_LAYER_the_world_fakesemfd
```

so no other process on the host sees a modified Vulkan environment. It builds with a single

`gcc -O2 -shared -fPIC` invocation against the Vulkan headers.

Why this does not invalidate the results.

The layer makes a capability appear present that is absent, which is exactly the kind of intervention that should be distrusted in a paper about whether evaluation results are determined by their inputs. Two things bound the risk. First, the mechanism: every path that would consume an fd-based semaphore is removed rather than faked, so the renderer takes the same code path it would take on a driver that had never advertised the extension — the layer does not substitute an approximate implementation for a real one. Second, and more convincing, the check in Sect. 6.2: this software-rendered, layer-mediated configuration reproduces the published success rates to a mean absolute difference of 0.026 across eight cells, and to +0.003 and −0.040 on the second embodiment. A renderer that differed materially from the reference would not agree to within a fraction of one seed’s worth of noise.

When it would be unsound.

If CUDA interop were enabled, the stubbed entry points would be reached and the synchronisation they are meant to provide would be absent. Any use of this layer with GPU interop active should be assumed incorrect until re-checked. Our runs use CPU physics with software rendering, so the path is never taken.

Appendix C Complete condition table with per-cell n

Appendix D Reproduction steps

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